

Motion along a curve

- 10 A particle's position is $\mathbf{r}(t) = \langle e^t, e^{-t}, \sqrt{2}t \rangle$. Find $\mathbf{v}(t)$ and show that the speed $|\mathbf{v}(t)|$ is minimized at $t = 0$.

Distance from a point to a line and plane

- 11 Find the distance from the point $P = (1, 2, 3)$ to the line through $(0, 0, 0)$ with direction $\mathbf{v} = \langle 1, 1, 1 \rangle$, using $d = \frac{|\mathbf{w} \times \mathbf{v}|}{|\mathbf{v}|}$ where $\mathbf{w} = \langle 1, 2, 3 \rangle$.
- 12 Find the distance from the point $(2, 1, -1)$ to the plane $x + 2y - 2z = 5$, using $d = \frac{|ax_0 + by_0 + cz_0 - d|}{\sqrt{a^2 + b^2 + c^2}}$.

Curvature and the normal component of acceleration

- 13 For a particle moving with position $\mathbf{r}(t) = \langle t, t^2, 0 \rangle$, find $\mathbf{v}(t) = \mathbf{r}'(t)$ and $\mathbf{a}(t) = \mathbf{r}''(t)$, then find the speed at $t = 1$.

Projections and angles between vectors

- 14 Find the angle between $\mathbf{a} = \langle 1, 1, 0 \rangle$ and $\mathbf{b} = \langle 1, 0, 1 \rangle$, using $\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$.
- 15 Find the scalar projection of $\mathbf{a} = \langle 3, 4 \rangle$ onto $\mathbf{b} = \langle 1, 0 \rangle$, using the formula $\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|}$.

An applied vector problem

- 16 An airplane's velocity relative to the air is $\mathbf{v}_{air} = \langle 200, 0 \rangle$ km/h (heading due east), and the wind velocity is $\mathbf{v}_{wind} = \langle 10, 30 \rangle$ km/h. Find the airplane's actual ground velocity $\mathbf{v}_{ground} = \mathbf{v}_{air} + \mathbf{v}_{wind}$, and find its ground speed.
- 17 A drone hovers with position vector $\mathbf{r}(t) = \langle 5\cos t, 5\sin t, 2 + 0.5t \rangle$ (meters, seconds), tracing a rising spiral. Find the drone's velocity vector $\mathbf{v}(t)$, then find the drone's speed at $t = \frac{\pi}{2}$, and separately find its height above the ground at that same instant.
- 18 For $\mathbf{a} = \langle 2, -3, 1 \rangle$ and $\mathbf{b} = \langle -1, 2, 4 \rangle$, find each of the following:
- a $2\mathbf{a} - \mathbf{b}$
 - b $\mathbf{a} \times \mathbf{b}$
 - c The area of the parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$

- 19 A particle's position is $\mathbf{r}(t) = \langle t^2 - 1, 2t, t^3 \rangle$. Find each of the following at $t = 1$:
- a The velocity vector $\mathbf{v}(1)$.
 - b The acceleration vector $\mathbf{a}(1)$.
 - c The speed $|\mathbf{v}(1)|$.